ATLANTIC RECORDING CORPORATION

US Playlist Analytics

Research Paper & Exploratory Data Analysis

27,800	943	297	555
Playlist Snapshots	Unique Songs	Unique Artists	Observation Days

Prepared for: Executive Leadership & Artist Strategy Team

Analysis Period: May 18, 2024 – November 27, 2025

Classification: Confidential & Proprietary

Executive Summary

This research paper presents a comprehensive exploratory data analysis (EDA) of the US Spotify Top 50 daily playlist over a 558-day observation period spanning May 2024 to November 2025. The dataset comprises 27,800 playlist snapshot records covering 943 unique songs across 297 artists, providing an unparalleled longitudinal view of music streaming dynamics in the United States.

The analysis focuses on two primary pillars: Popularity Analytics and Ranking Trends & Volatility. The findings reveal nuanced patterns in how songs achieve, maintain, and lose playlist prominence — insights that directly inform Atlantic Recording Corporation's artist promotion strategies, release timing decisions, and marketing spend optimization.

Key Findings at a Glance

- Longevity leaders dominate: 'Something in the Orange' by Zach Bryan held the chart for 536 days — nearly the entire observation period — demonstrating that country and folk-adjacent tracks achieve exceptional persistence.
- Popularity-rank correlation is weak ($r = -0.09$): High popularity scores do not reliably predict top-10 rank positions, suggesting the playlist algorithm weighs streaming velocity and editorial curation heavily alongside raw popularity.
- Singles outperform albums: Tracks released as singles achieve an average popularity score of 90.1 vs. 86.7 for album tracks, and rank nearly 2 positions higher on average.
- Taylor Swift dominates artist presence: With 2,016 chart appearances across 96 unique songs — nearly double her closest competitor — Swift's catalog strategy is the gold standard for playlist dominance.
- Explicit content carries minimal disadvantage: Explicit tracks average 87.3 vs. 88.0 for clean tracks — a negligible gap that should not drive release decisions.
- Optimal song duration is 4–5 minutes: Songs in this range achieve the best average chart position (rank 23.4), outperforming shorter and longer tracks.

1. Introduction & Research Objectives

The streaming era has fundamentally transformed how music achieves commercial success. Unlike the album-sales model of previous decades, sustained chart presence on curated editorial playlists directly drives streams, revenue, and artist visibility. The Spotify Top 50 daily playlist represents one of the highest-impact touchpoints in the US market, with placement capable of generating millions of incremental streams within 24–48 hours of entry.

Despite access to rich daily playlist data, stakeholders at Atlantic Recording Corporation have historically relied on anecdotal observations rather than systematic analytics. This research bridges that gap by applying rigorous exploratory data analysis to 555 days of continuous playlist observation.

1.1 Research Objectives

- Quantify ranking stability and volatility across the full chart universe.
- Identify songs and artists with exceptional longevity and characterize their attributes.
- Examine the relationship between Spotify popularity scores and playlist rank.
- Analyze how content attributes (explicit content, album type, duration) relate to chart performance.
- Deliver actionable strategic recommendations for label operations.

1.2 Dataset Overview

Field	Type	Description	Coverage
date	Date	Daily playlist snapshot date	555 unique dates
position	Integer	Chart rank (1–50)	Fully populated, 1–50
song	String	Track title	943 unique songs
artist	String	Artist name(s)	297 unique artists
popularity	Integer	Spotify popularity score (0–100)	Mean: 87.67
duration_ms	Integer	Track length in milliseconds	~2–8 min range
album_type	Categorical	Single / Album / Compilation	3 categories
total_tracks	Integer	Tracks in source album	1 to 36+
is_explicit	Boolean	Explicit content flag	48% explicit

2. Data Ingestion & Quality Validation

2.1 Data Quality Assessment

The dataset was subjected to comprehensive validation checks prior to analysis. The results confirm a high-integrity dataset suitable for rigorous statistical analysis.

Missing values across all fields	**0 / 27,800**	Clean — fully complete
Duplicate song-date entries detected	**0**	Verified — no duplicates
Rank range compliance (positions 1–50)	**100%**	All entries valid
Observation coverage	**555 dates**	May 2024 – Nov 2025

2.2 Feature Engineering

The following derived metrics were computed at the song level to enable longitudinal performance analysis:

Derived Metric	Formula / Method	Purpose
Days on Chart	Count of unique dates per song	Longevity indicator
Average Rank	Mean of daily position values	Overall performance index
Best Rank Achieved	Minimum position value (lower = better)	Peak success metric
Rank Volatility Index	Standard deviation of daily position	Stability measurement
Duration (minutes)	duration_ms / 60,000	Content attribute analysis
Popularity Trend Score	Average popularity over chart presence	Listener engagement proxy

3. Ranking Trends & Volatility Analysis

Chart position is the primary currency of playlist performance. This section examines how songs move through the Top 50, distinguishing songs that achieve sustained stable positions from those that spike and exit quickly. The Rank Volatility Index (RVI), measured as the standard deviation of daily chart position, serves as the core stability metric throughout this section.

3.1 Chart Longevity — Songs with Greatest Playlist Persistence

The following songs achieved the highest number of unique days on the Top 50 playlist. These represent the benchmark for catalog resilience and sustained audience engagement.

#	Song	Artist	Days	Avg Pos	Best	Avg Pop
1	Something in the Orange	Zach Bryan	536	26.3	#9	88.4
2	Last Night	Morgan Wallen	492	22.6	#1	85.7
3	I Remember Everything (feat. Kacey …	Zach Bryan	446	13.9	#1	90.1
4	See You Again (feat. Kali Uchis)	Tyler, The Creator	444	29.4	#4	91.1
5	Stick Season	Noah Kahan	414	18.3	#1	91.1
6	Cruel Summer	Taylor Swift	388	16.9	#1	97.0
7	Lose Control	Teddy Swims	329	23.5	#5	90.9
8	Thinkin' Bout Me	Morgan Wallen	324	39.0	#21	86.0
9	Beautiful Things	Benson Boone	307	14.7	#1	90.9
10	You Proof	Morgan Wallen	302	32.6	#9	84.7

Strategic Insight: Zach Bryan places two songs in the top 3 longevity leaders (combined 982 days). Morgan Wallen contributes three songs to the top 10, confirming that country and Americana genres drive exceptional chart persistence in the US market — a compelling finding that should inform genre investment strategy.

3.2 Rank Stability Analysis — Volatility Index

A low RVI indicates consistent day-to-day rank positioning; a high RVI signals dramatic rank swings. Songs with fewer than 30 days on chart were excluded to ensure statistical reliability.

3.2.1 Most Stable Songs (Lowest RVI — min. 30 days on chart)

Song	Artist	RVI Score	Days	Avg Position
Que Onda	Calle 24 & Chino Pacas & …	2.92	56	44.6

Song	Artist	RVI Score	Days	Avg Position
Shut up My Moms Calling	Hotel Ugly	2.94	45	46.4
BIRDS OF A FEATHER	Billie Eilish	3.12	191	5.3
Si Antes Te Hubiera Conocido	KAROL G	3.31	51	46.6
Bad Habit	Steve Lacy	3.45	34	44.1
Die With A Smile	Lady Gaga & Bruno Mars	3.63	103	3.5
Calm Down (with Selena Gomez)	Rema	3.71	52	42.6
LA PATRULLA	Peso Pluma	3.71	44	41.1

3.2.2 Most Volatile Songs (Highest RVI — min. 10 days on chart)

Song	Artist	RVI Score	Days on Chart
Back To December (Taylor's Version)	Taylor Swift	19.08	11
You're Losing Me (From The Vault)	Taylor Swift	18.61	16
BACK TO ME	Kanye West & Ty Dolla $ig…	18.54	11
3D (feat. Jack Harlow)	Jung Kook & Jack Harlow	18.43	12
I Can See You (Taylor's Version) (From T…	Taylor Swift	17.97	36
Is It Over Now? (Taylor's Version) (From…	Taylor Swift	17.87	89
CARNIVAL	Kanye West & Ty Dolla $ig…	17.76	107
Enchanted (Taylor's Version)	Taylor Swift	17.00	11

Strategic Insight: High volatility strongly correlates with vault releases and album-era re-entries. Taylor Swift's vault tracks exhibit the highest RVI scores — they surge onto the chart with album releases, then dissipate rapidly. This pattern suggests re-release campaigns generate short, intense chart bursts rather than sustained presence.

3.3 Chart Entry & Exit Behavior

Across 943 tracked songs, the average chart lifespan was 29.4 days. The distribution is heavily right-skewed — approximately 70% of songs appear on the chart for fewer than 30 days, while the top longevity leaders exceeded 300+ days. Sustained chart presence is an exceptional achievement, not a baseline outcome.

Longevity Segment	Est. Share of Songs	Characteristics
1–7 days (Flash entries)	~40%	Short-lived spikes; release-week or viral moment driven

Longevity Segment	Est. Share of Songs	Characteristics
8–30 days (Short-term)	~30%	Standard chart cycle; moderate listener retention
31–90 days (Medium-term)	~20%	Strong adoption; active promotion period
90+ days (Chart anchors)	~10%	Cultural resonance; catalog classics in formation

4. Popularity Score Analytics

The Spotify popularity score is a 0–100 index reflecting recent streaming activity. It is distinct from chart rank in that it captures cross-platform and off-playlist streaming — a song can be highly popular without currently charting in the Top 50, and vice versa.

4.1 Overall Popularity Distribution

Metric	Value	Interpretation
Dataset Mean Popularity	87.67 / 100	Exceptionally high — Top 50 self-selects for popularity
Median Popularity	89.0 / 100	Over half of chart entries exceed score of 89
Standard Deviation	10.46	Moderate spread within a high-value range
Minimum Observed	0	Likely older catalog tracks with minimal recent streaming
Maximum Observed	100	Multiple tracks achieved peak popularity score
25th Percentile	85.0	Even lowest quartile maintains strong engagement
75th Percentile	93.0	Top quartile clusters near the popularity ceiling

4.2 Popularity by Chart Tier

A central analytical question is whether chart rank and popularity score are tightly correlated — do the most popular songs consistently dominate top positions? The data reveals a more nuanced picture.

Chart Tier	Avg Popularity Score	Delta vs Lower Tier
Top 10 (Ranks 1–10)	89.62	+2.01 vs Ranks 11–20
Upper Mid (Ranks 11–20)	87.60	+0.57 vs Ranks 21–50
Lower Half (Ranks 21–50)	87.04	Baseline tier

The popularity gradient across chart tiers is surprisingly shallow — the Top 10 averages only 2.6 points higher than the bottom half. This means rank is not primarily determined by popularity score alone. Streaming velocity, editorial curation decisions, and playlist algorithm weighting all play significant independent roles.

4.3 Popularity–Rank Correlation

Pearson Correlation: Popularity Score vs Chart Position	**r = -0.095**	Weak inverse relationship

The Pearson correlation coefficient of -0.095 (negative because lower rank numbers represent better positions) has significant strategic implications:

- Chart position cannot be reliably predicted from popularity score alone.
- A song with popularity 85 is nearly as likely to chart in the Top 5 as a song with popularity 95, given effective release timing and promotion.
- Investment in streaming volume (popularity) should be paired with editorial playlist pitching, as neither alone guarantees top-tier placement.
- Popularity may better predict revenue sustainability — a high-popularity, mid-ranked song likely generates more cumulative streams than a low-popularity, momentarily top-ranked song.

4.4 Songs with Highest Average Popularity Score

Songs with fewer than 7 days on chart were excluded to avoid single-day anomalies. These tracks represent the most listener-engaged songs throughout the observation period.

Song	Artist	Avg Popularity	Days on Chart	Best Rank
LALA	Myke Towers	98.7	49	#29
Cupid - Twin Ver.	FIFTY FIFTY	98.2	71	#7
La Bebe - Remix	Yng Lvcas & Peso Pluma	98.0	113	#6
BIRDS OF A FEATHER	Billie Eilish	97.2	191	#2
Cruel Summer	Taylor Swift	97.0	388	#1
Die With A Smile	Lady Gaga & Bruno Mars	96.8	103	#1
La Diabla	Xavi	96.7	104	#5
Seven (feat. Latto) (Explicit Ver.)	Jung Kook & Latto	96.3	89	#1
i like the way you kiss me	Artemas	96.2	100	#3
Gata Only	FloyyMenor & Cris Mj	96.2	113	#13

Strategic Insight: Latin and K-pop tracks feature prominently in the highest-popularity segment, reflecting global streaming fan bases outside the US that contribute to popularity scores without necessarily influencing the US chart position directly. This creates a dual-track opportunity — international artist promotion can generate exceptional popularity metrics that benefit catalog valuation even when US chart impact is moderate.

5. Artist Performance & Dominance Analysis

Artist dominance is measured as total playlist appearances (song-days), reflecting both catalog breadth and sustained audience engagement. This metric rewards artists who maintain multiple songs on the chart simultaneously or achieve prolonged single-song presence.

5.1 Artist Dominance Leaderboard

#	Artist	Total Appearances	Unique Songs	Avg Popularity	Best Rank
1	Taylor Swift	2,016	96	87.1	#1
2	Zach Bryan	1,860	35	86.0	#1
3	Morgan Wallen	1,672	10	85.4	#1
4	Sabrina Carpenter	968	14	90.2	#1
5	Drake	844	35	84.7	#1
6	Chappell Roan	794	6	88.7	#1
7	Travis Scott	764	31	88.6	#1
8	Billie Eilish	729	12	92.8	#2
9	Tyler, The Creator	706	16	88.4	#1
10	SZA	650	3	91.7	#3

5.2 Catalog Depth vs. Concentration Strategy

Two contrasting approaches to chart dominance emerge from the data, both capable of generating substantial total appearances:

Artist	Songs	Total Appearances	Strategy Profile
Taylor Swift	96	2,016	Broad catalog depth — re-release driven
Zach Bryan	35	1,860	Balanced breadth — organic longevity
Morgan Wallen	10	1,672	Concentrated hits — fewer, longer-running songs
Sabrina Carpenter	14	968	Rising star — concentrated breakout period
SZA	3	650	Hyper-concentrated — extreme song-level virality

Taylor Swift's 96-song breadth is a direct result of the Taylor's Version re-release strategy, which reactivates legacy catalog tracks and multiplies chart opportunities simultaneously. Morgan Wallen achieves comparable dominance with just 10 songs — demonstrating that extreme hit concentration can rival catalog breadth in total appearance count. Both strategies are viable; the right approach depends on the artist's catalog maturity and audience loyalty.

6. Content Attribute Analysis

6.1 Explicit vs. Non-Explicit Content Performance

Explicit content constitutes 48% of all playlist entries — a near-even split that reflects the genre diversity of the Top 50. The performance comparison between explicit and clean tracks reveals minimal differentiation.

Content Type	Share of Chart	Avg Popularity	Avg Chart Position
Non-Explicit (Clean)	52.0%	88.0	25.6
Explicit	48.0%	87.3	25.4
Difference	—	-0.7	-0.2 (explicit ranks marginally better)

Strategic Recommendation: Explicit content labeling should not be a gating factor in chart promotion decisions. The performance gap is statistically negligible. Artist creative vision should take precedence over content classification in release strategy.

6.2 Album Type Comparison

Songs are categorized into three release types: Singles (stand-alone releases), Albums (tracks within multi-song projects), and Compilations. Performance varies meaningfully across these categories.

Album Type	Chart Share	Avg Popularity Score	Avg Chart Position
Single	28.8%	90.1	24.0
Album	71.0%	86.7	26.1
Compilation	0.2%	78.4	33.2

Singles consistently outperform album tracks across both popularity score (+3.4 points) and chart position (+2.1 positions on average). This likely reflects the focused marketing attention and playlist pitching resources directed at single releases, combined with the algorithmic boost Spotify provides to newly released stand-alone tracks. Compilation tracks underperform significantly on both metrics.

6.3 Song Duration Impact Analysis

Song duration is a frequently debated factor in streaming strategy. The data provides clear guidance on the optimal length range for chart performance.

Duration Range	Avg Popularity Score	Avg Chart Position	Interpretation

Duration Range	Avg Popularity Score	Avg Chart Position	Interpretation
< 2 minutes	85.3	31.6	Short tracks underperform — novelty, limited replay
2–3 minutes	88.0	25.4	Strong — classic pop/radio length
3–4 minutes	87.9	25.9	Strong — standard song structure
4–5 minutes	87.0	23.4	Best chart position — extended engagement
> 5 minutes	83.9	25.3	Declining returns — listener fatigue risk

Strategic Recommendation: Songs in the 4–5 minute range achieve the best average chart position (23.4), outperforming the popular 2–3 minute range by over 2 positions. Very short tracks (<2 min) significantly underperform. Artists should resist pressure to shorten songs purely for algorithmic reasons, as medium-length tracks demonstrate superior chart staying power.

7. Key Performance Indicators Summary

The following KPI dashboard consolidates the most critical metrics derived from the analysis into a single reference view for executive stakeholders.

KPI	Value	Context
Days on Chart (Avg)	29.4 days	Baseline longevity across all 943 songs
Days on Chart (Max)	536 days	'Something in the Orange' — Zach Bryan
Average Rank (All Songs)	25.5	Uniform distribution across positions 1–50
Avg Rank — Top Longevity Songs	19.8	Long-lasting songs cluster in upper half
Popularity–Rank Correlation	r = -0.095	Weak — chart rank ≠ popularity score
Avg Popularity — Top 10	89.6 / 100	2.6 pts above lower-half average
Avg Popularity — All Chart	87.7 / 100	High floor — Top 50 self-selects strongly
Explicit Content Share	48%	Near-even split; negligible performance gap
Single vs Album Popularity Gap	+3.4 pts (singles)	Singles outperform album tracks consistently
Optimal Song Duration	4–5 minutes	Best avg chart position: rank 23.4
Artist Dominance Leader	Taylor Swift	2,016 appearances, 96 unique songs
Most Stable Song (RVI)	BIRDS OF A FEATHER — 3.12	191 days, avg rank 5.3
Most Volatile Song (RVI)	Back To December (TV) — 19.1	11 days, extreme rank swings

8. Strategic Recommendations

The following recommendations are derived directly from analytical findings and are designed to be actionable within Atlantic Recording Corporation's existing operational structure.

8.1 Artist & Catalog Strategy

- Invest in country and folk-adjacent artists for long-term chart presence. The genre consistently produces the longest-duration chart entries in the dataset, with Zach Bryan and Morgan Wallen combining for over 3,500 total chart appearances.
- Replicate catalog re-release strategy for back-catalog artists. Taylor Swift's Vault approach demonstrates that archival tracks can generate substantial new chart activity when marketed as fresh releases.
- Identify 'anchor songs' — tracks with low RVI scores and 90+ days on chart — and sustain promotional investment around these songs, as they generate compounding streams over time.

8.2 Release & Promotion Strategy

- Prioritize single releases for new artists targeting immediate chart impact. Singles average 3.4 popularity points and 2.1 chart positions above album tracks.
- Do not curtail songs to under 2 minutes for streaming optimization. Very short tracks significantly underperform on both popularity and chart position. Target the 3–5 minute sweet spot.
- Decouple popularity score tracking from chart rank monitoring — high popularity does not reliably translate to high rank, and both require independent optimization strategies.

8.3 Content & Marketing Strategy

- Remove explicit content restrictions from playlist promotion decisions. The performance gap between explicit and clean tracks is 0.7 popularity points — statistically negligible and insufficient to justify artistic compromise.
- Invest in editorial playlist relationships independent of popularity score. The weak correlation ($r = -0.095$) confirms that curation access is a distinct lever from streaming volume.
- Leverage international fan bases (Latin, K-pop) to drive Spotify popularity scores as a catalog valuation strategy, even when US chart impact is secondary.

9. Conclusion

This research paper has delivered a comprehensive, data-driven analysis of 555 days of US Spotify Top 50 playlist dynamics, covering 943 songs, 297 artists, and 27,800 daily snapshot records. The analysis reveals that playlist success is a multi-dimensional phenomenon — one where chart rank, popularity score, content type, and song duration each play distinct and partially independent roles.

The most significant finding is the weakness of the popularity–rank correlation ($r = -0.095$), which fundamentally challenges the assumption that driving streaming volume alone is sufficient for chart success. Editorial relationships, release timing, genre selection, and catalog strategy are equally critical levers.

Country and Americana genres demonstrate outsized chart persistence relative to their streaming market share. Singles outperform album tracks. Explicit content carries no meaningful penalty. Songs of 4–5 minutes duration achieve the best chart positions. These findings, grounded in 18 months of real playlist behavior, provide Atlantic Recording Corporation with a clear, evidence-based framework for strategic decision-making in the streaming era.

Finding	Implication for Atlantic RC
Country/Americana = longest chart presence	Increase genre investment and promotion budget allocation
Singles outperform album tracks by 3.4 pts	Prioritize single-first release strategies for new artists
Popularity ≠ Rank ($r = -0.095$)	Maintain separate editorial pitching and streaming growth teams
Explicit content: negligible performance gap	Eliminate explicit content as a promotion gating criterion
4–5 min songs rank best on average	Resist artificial song shortening for algorithmic optimization
Catalog re-releases generate burst activity	Develop systematic vault/re-release pipeline for back-catalog

— End of Report —